CLINICAL TRIAL FAILURES · DISCONTINUATION RATE · ONCOLOGY PHASE 2,3 · STARTS 2015–2024

HDAC: 3 molecules stopped early, all small molecules

4 stopped trials, but 3 molecules — all small molecules: Tucidinostat, Pracinostat and Vorinostat. Against 5.2% for all oncology trials with a resolved drug target and 4.5% across all oncology Phase 2,3 trials in the same window. The rate counts trials that stopped early for efficacy, safety or benefit–risk reasons. Trials that ran to completion and missed their primary endpoint are counted separately below, never in the rate.

5.8%
4 of 69 closed trials (95% CI 2.3%–14.0%)

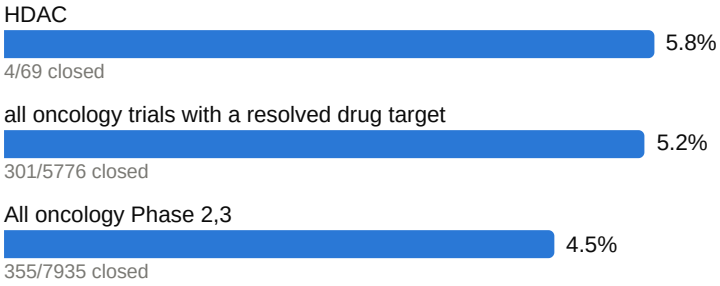
3 / 0 / 0
efficacy / safety / both (adds to 4)

31.8%
of 217 trials have closed · 148 still open or unresolved

1.8%
stopped for a biological reason within 3 years of starting (95% CI 0.5%–7.2%; 74 still at risk; 1.5% area-wide)

1 of 1 **Completed and missed the primary endpoint.** Of 46 completed trials in this segment, 1 posted a primary result on ClinicalTrials.gov that can be read — the sponsor's own comparison held to the sponsor's own threshold, or its statement that the endpoint was missed: 1 missed and 0 met. Counted separately and never in the rate above; a floor, not a rate. 22 of the 45 completed trials that finished more than a year ago have posted no results at all — a gap in what can be known, not a sign of failure.

How this compares



What is in this class: 162 distinct experimental drugs across 217 trials (small molecule 214, antibody 62, unknown 37, protein 30). Most tested: TUCIDINOSTAT (128), CYCLOPHOSPHAMIDE ANHYDROUS (18), VORINOSTAT (16), ENTINOSTAT (16), DEXAMETHASONE (15). A class groups drugs by what they act on; check that the grouping is one you would make before reading the rate as a property of the mechanism.

How much does this rest on one decision? The 4 stops came from 4 sponsor-asset programmes across 4 sponsors; the largest (University of California, San Francisco) contributed 1; removing that programme's trials from both sides leaves 4.5%. Screened alongside 85 other segments, the rate is within what screening that many classes could produce (q=1); the molecules and trials that failed here are the record either way. A further 5 closed trials here were terminated with no cause recorded in the registry; if every one of them were biological the rate would be 13.0%. Read it as a screen worth checking against the underlying trials.

Bars show the share of closed trials stopped for biological reasons. Confidence intervals overlap where sample sizes are small — read the counts, not just the bars.

By start cohort

Trials started	Stops	Closed	Rate	95% CI
2015–2017	2	34	5.9%	1.6%–19.1%
2018–2020	1	23	4.3%	0.8%–21.0%
2021–2024	1	12	8.3%	1.5%–35.4%

Where the stops sit

Efficacy / futility: 3, Benefit–risk: 1.
Sponsors with most stops: Great Novel Therapeutics Biotech & Medicals Corporation (1), Sun Yat-sen University (1), Helsinn Healthcare SA (1), University of California, San Francisco (1).

The stopped trials

Trial	Ph	Sponsor	Experimental drugs	Stopped	Type	Registry stop reason
NCT05770882	2	Great Novel Therapeutics Biotech & Medicals Corporation	TUCIDINOSTAT, REGORAFENIB	2025-01	Benefit–risk	The overall profile does not support development for Hepatocellular Carcinoma
NCT04651127	2	Sun Yat-sen University	TORIPALIMAB, TUCIDINOSTAT	2023-01	Efficacy	Five patients were enrolled in the study, but no treatment response was observed and all the patients had the best of response of prgressive disease. Given the lack of efficacy, the trial was terminated.
NCT03151408	3	Helsinn Healthcare SA	PRACINOSTAT	2020-08	Efficacy	The IDMC recommended to stop the study prematurely due to a lack of efficacy.
NCT02395627	2	University of California, San Francisco	TAMOXIFEN, VORINOSTAT, PEMBROLIZUMAB	2019-06	Efficacy	Insufficient efficacy in an unselected patient population

Method

Denominator: ClinicalTrials.gov interventional Phase 2,3 oncology trials started 2015–2024 that have closed (completed or terminated). Numerator: terminated trials whose registry stop reason is classified as biological (efficacy, safety or benefit–risk) by Classification V2 — held-out precision 95.5%, recall 95.3% (n=600). Drugs are linked to ChEMBL and the NCI Thesaurus; 76% of industry oncology trials in this window carry a resolved drug target, and a trial without one cannot enter a mechanism class. Intervals are Wilson 95%.

Limits

Not a failure rate: completed trials that missed their endpoints are not in the rate. They are counted separately where the sponsor posted a primary result that can be read — most did not, so that count is a floor — and programmes discontinued after a completed trial do not appear. Stop reasons are sponsor-reported. This is a closed-trial proportion, not a time-to-event analysis: only 31.8% of trials in this segment have closed, and a trial that stops early enters the denominator sooner than one that runs to completion, which can inflate the rate in immature segments. Recent cohorts have fewer closed trials, so their rates are less stable. Research signals, not clinical or investment advice.

The whole cohort, trial by trial. The diligence report names every trial behind these figures — the ones still running and the ones that completed and missed their primary endpoint included — and places your molecule against each one that failed. Free during the beta, in exchange for a work email, delivered immediately: **clinicaltrialfailures.com/asset-check**

Sources: ClinicalTrials.gov (NLM); ChEMBL (EMBL-EBI, CC BY-SA 3.0); NCI Thesaurus (NCI); RxNorm/RxClass (NLM); SEC EDGAR. Classification, linkage and rates by Clinical Trial Failures. Rebuilt weekly; this brief covers trials started 2015–2024.